

Exercise and worked solution

The economics of a mandate

This exercise revisits, with figures, the economics of a mandate seen in module 1: what fees cost the client, what they earn the manager, and why the fee level decides almost everything. Work through each part before reading the solution.

Part A · The cost to the client

PROBLEM

A client places €250,000 under management. The mandate targets a 7% gross annual return, with a 90 bps fee charged each year on the value. Horizon: 15 years. Compute (a) the net final value, (b) the total fees actually paid, (c) the value reached with no fees, and (d) the shortfall. Comment on the gap between total fees and shortfall.

SOLUTION

1. Each year the value grows 7% then loses 0.90% to fees. The net annual factor is $1.07 \times (1 - 0.0090) = 1.0604$.
2. (a) Net final value: $€250,000 \times 1.0604^{15} \approx \text{€602,284}$. Net annualised return $\approx 6.04\%$.
3. (b) Total fees actually paid (sum of yearly charges) $\approx \text{€56,195}$.
4. (c) With no fees: $€250,000 \times 1.07^{15} \approx \text{€689,758}$.
5. (d) Shortfall = $€689,758 - €602,284 \approx \text{€87,474}$.
6. Comment: the shortfall (€87,474) far exceeds the fees paid (€56,195). The difference is the lost compounding: every euro taken out stops working in later years.

Part B · The revenue to the manager

PROBLEM

A management company runs a €400m fund at 85 bps of management fee. The fund also charges a performance fee of 15% of the outperformance above a 5% hurdle, subject to a high-water mark (the fund is at its peak at the start of the year). This year the gross return reaches 11%. Compute the management revenue, the performance fee, the total revenue, then express that total in bps of AuM.

SOLUTION

1. Management revenue = $\text{AuM} \times 0.85\% = €400\text{m} \times 0.0085 = \text{€3.4m}$.
2. Outperformance above the hurdle = $11\% - 5\% = 6\%$. The high-water mark holds (the fund starts at its peak), so the fee is due.
3. Performance fee = $15\% \times 6\% \times \text{AuM} = 0.90\% \times €400\text{m} = \text{€3.6m}$.
4. Total revenue = $€3.4\text{m} + €3.6\text{m} = \text{€7m}$.
5. In bps of AuM: $€7\text{m} / €400\text{m} = \text{175 bps}$. In a good year, the performance fee nearly doubles the manager's revenue.

Part C · Fee pressure

PROBLEM

A boutique wants to earn €5m of management revenue per year. (a) What AuM does it need at 80 bps? (b) And if ETF competition brings its average fee down to 55 bps? (c) By how much must its AuM grow to compensate?

SOLUTION

1. (a) $\text{AuM} = \text{revenue} / \text{fee} = €5\text{m} / 0.0080 = \text{€625m}$.
2. (b) At 55 bps: $€5\text{m} / 0.0055 \approx \text{€909.1m}$.
3. (c) Required growth $\approx €284.1\text{m}$, i.e. **+45.5%** of AuM for the same revenue.
4. Comment: when fees fall, far more assets must be gathered to hold the same revenue. This is the engine of the industry's race for scale.

Part D · Active versus passive, over 10 years

PROBLEM

Two funds target the same 6% gross annual return over 10 years, on €100,000 invested. Active fund: TER 1.10%. Index ETF: TER 0.20%. Give the final value of each, then the gap in euros and as a percentage of the starting capital.

SOLUTION

1. Net annual factor = $1.06 \times (1 - \text{TER})$. Active: $1.06 \times 0.9890 = 1.0483$. Passive: $1.06 \times 0.9980 = 1.0579$.
2. Active fund: $€100,000 \times 1.0483^{10} \approx \mathbf{€160,332}$.
3. Index ETF: $€100,000 \times 1.0579^{10} \approx \mathbf{€175,535}$.
4. Gap $\approx \mathbf{€15,203}$, i.e. $\approx \mathbf{15.2\%}$ of the starting capital.
5. Comment: with no difference in skill, the fee gap alone creates a 15% wealth gap over ten years. That is why active management must beat its index by at least the fee gap, every year, just to break even.

Key takeaway: the fee level is the most reliable variable in the whole equation. The manager maximises it under a fiduciary constraint; the informed client minimises it.